Perimeters and areas of similar figures

ANSWER KEY

Core Concepts: Perimeters and areas of similar figures

1 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 140°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (40^\circ + 45^\circ) = 95^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 10

c. 8 ✓

d. 13

Substituting $x = 1$ into $y = 5x + 3$ gives $y = 5(1) + 3 = 8$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 140°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (40^\circ + 52^\circ) = 88^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 8

b. 15

c. 13 ✓

d. 18

Substituting $x = 2$ into $y = 5x + 3$ gives $y = 5(2) + 3 = 13$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 140°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (40^\circ + 59^\circ) = 81^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 13

b. 20

c. 18 ✓

d. 23

Substituting $x = 3$ into $y = 5x + 3$ gives $y = 5(3) + 3 = 18$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying perimeters and areas of similar figures, $m\angle A = 40^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 74° ✓

b. 84°

c. 66°

d. 140°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (40^\circ + 66^\circ) = 74^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm ✓

c. 24 cm

d. 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 18

b. 25

c. 23 ✓

d. 28

Substituting $x = 4$ into $y = 5x + 3$ gives $y = 5(4) + 3 = 23$.

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